

Assignment 2

Group 11

BIG DATA TOOLS & TECHNIQUES

Kirichenko, Gavriel

Arul, Callum Joseph

Bhattarai, Diparshan

Chang, Chia-Wei

Giri, Bikash

Hussainyar, Friba

Liu, Hsi-Teng

Medrano Diaz, Sergio Alejandro

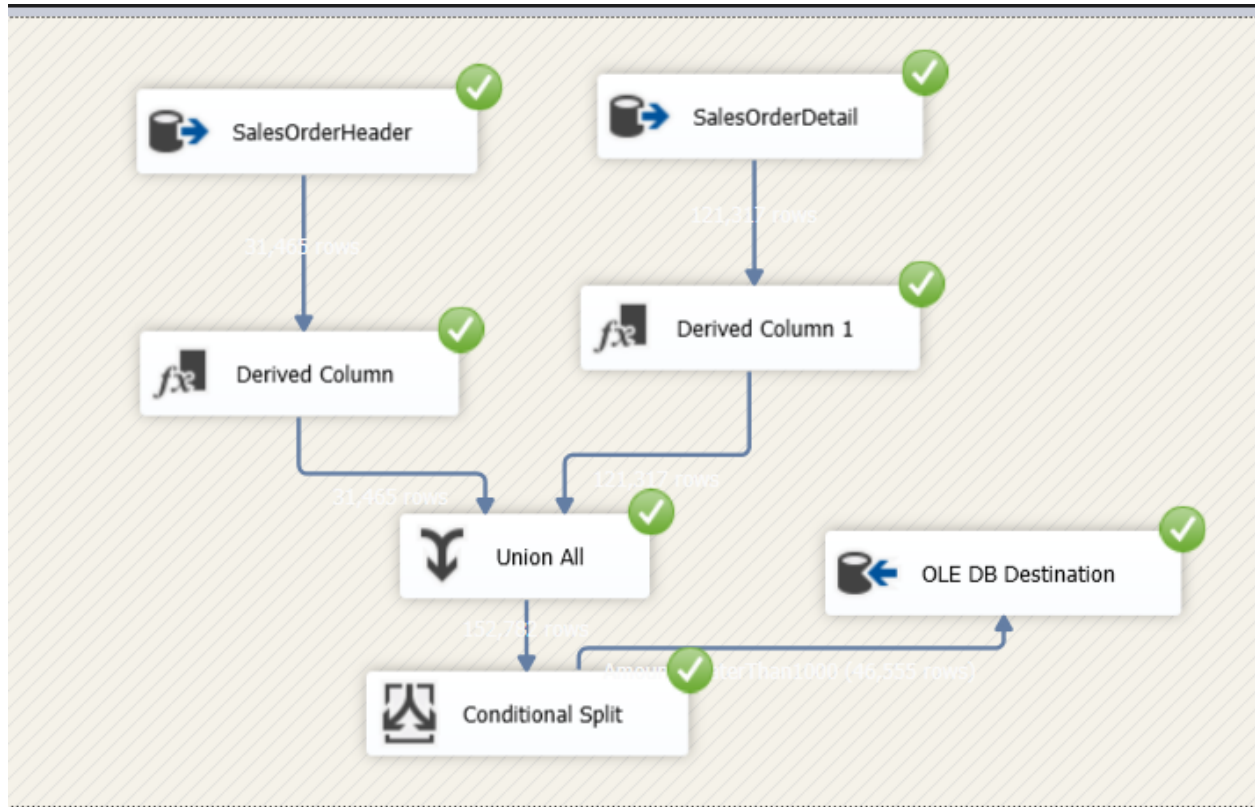
Omisesan, Abdul-Rasaq

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1. SSIS Pipeline

1.1. Data Flow Layout



1.2. Components

1.2.1. OLE DB Source

OLE DB Source Editor

Configure the properties used by a data flow to obtain data from any OLE DB provider.

Connection Manager
Columns
Error Output

Specify an OLE DB connection manager, a data source, or a data source view, and select the data access mode. If using the SQL command access mode, specify the SQL command either by typing the query or by using Query Builder.

OLE DB connection manager:
DESKTOP-S19DI25\SQLEXPRESS New...

Data access mode:
SQL command

SQL command text:
SELECT SalesOrderID, OrderDate, TotalDue
FROM Sales.SalesOrderHeader

Parameters...
Build Query...
Browse...
Parse Query

Preview...

OK Cancel Help

OLE DB Source Editor

Configure the properties used by a data flow to obtain data from any OLE DB provider.

Connection Manager
Columns
Error Output

Specify an OLE DB connection manager, a data source, or a data source view, and select the data access mode. If using the SQL command access mode, specify the SQL command either by typing the query or by using Query Builder.

OLE DB connection manager:
DESKTOP-S19DI25\SQLEXPRESS New...

Data access mode:
SQL command

SQL command text:
SELECT SalesOrderID, ModifiedDate AS OrderDate, LineTotal
FROM Sales.SalesOrderDetail

Parameters...
Build Query...
Browse...
Parse Query

Preview...

OK Cancel Help

1.2.2. Derived Columns

Derived Column Transformation Editor

Specify the expressions used to create new column values, and indicate whether the values update existing columns or populate new columns.

Variables and Parameters

Columns

Mathematical Functions

String Functions

Date/Time Functions

NULL Functions

Type Casts

Operators

Description:

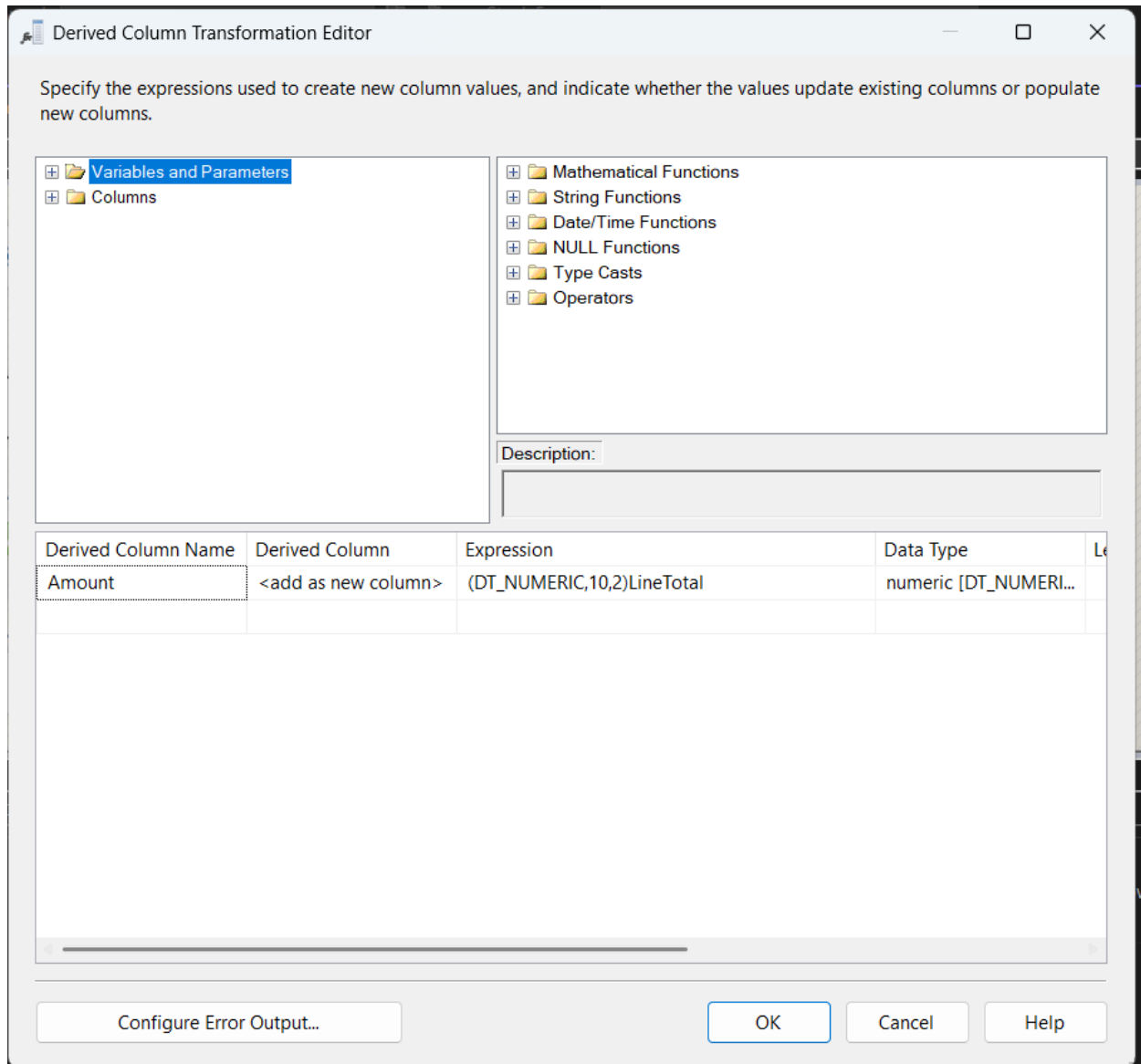
Derived Column Name	Derived Column	Expression	Data Type	Label
Amount	<add as new column>	(DT_NUMERIC,10,2)TotalDue	numeric [DT_NUMERI...	

Configure Error Output...

OK

Cancel

Help



1.2.3. Union All

Union All Transformation Editor

Configure the properties used to merge multiple inputs into one output by creating mappings between columns.

Output Column Name	Union All Input 1	Union All Input 2
SalesOrderID	SalesOrderID	SalesOrderID
OrderDate	OrderDate	OrderDate
Amount	Amount	Amount

OK Cancel Help

1.2.4. Conditional Split

Conditional Split Transformation Editor

Specify the conditions used to direct input rows to specific outputs. If an input row matches no condition, the row is directed to a default output.

+ Variables and Parameters

+ Columns

+ Mathematical Functions

+ String Functions

+ Date/Time Functions

+ NULL Functions

+ Type Casts

+ Operators

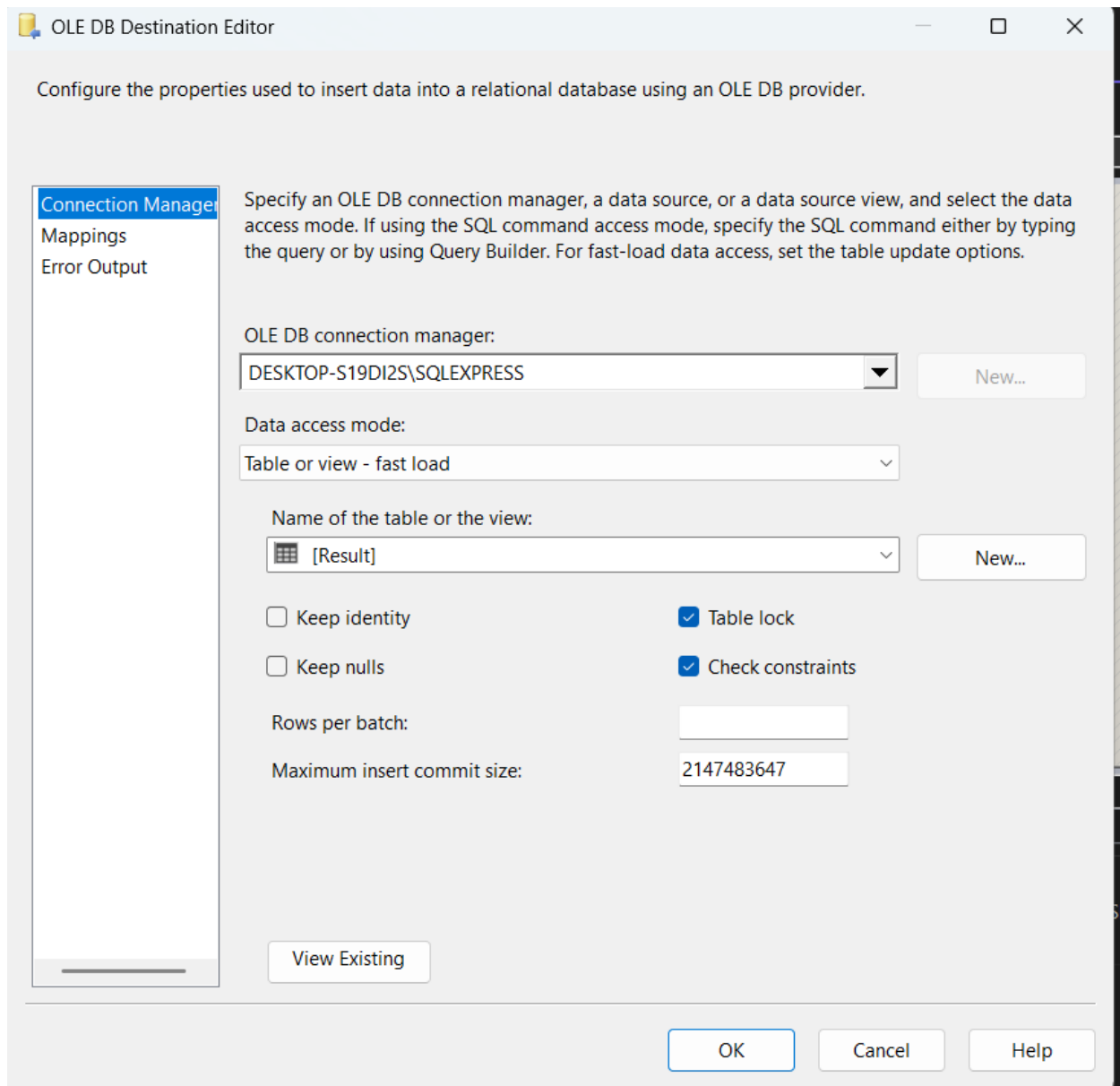
Description:

Order	Output Name	Condition
1	AmountGreaterThan1000	Amount > 1000

Default output name: Conditional Split Default Output

Configure Error Output... OK Cancel Help

1.2.5. OLE DB Destination



The image shows the 'OLE DB Destination Editor' dialog box. It has a title bar with a yellow icon and the text 'OLE DB Destination Editor'. Below the title bar is a subtitle: 'Configure the properties used to insert data into a relational database using an OLE DB provider.' On the left is a sidebar with three items: 'Connection Manager' (highlighted in blue), 'Mappings', and 'Error Output'. The main area contains the following fields and controls:

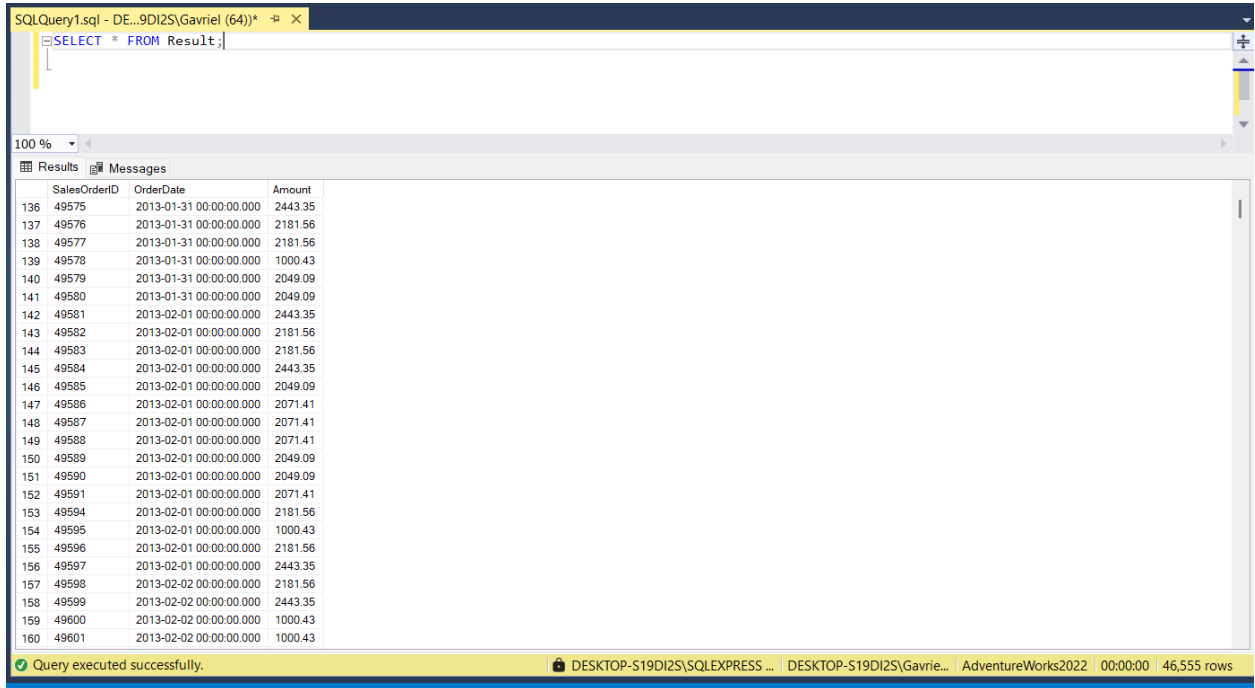
- OLE DB connection manager:** A dropdown menu showing 'DESKTOP-S19DI2S\SQLEXPRESS' and a 'New...' button to its right.
- Data access mode:** A dropdown menu showing 'Table or view - fast load'.
- Name of the table or the view:** A dropdown menu showing '[Result]' and a 'New...' button to its right.
- Keep identity:** An unchecked checkbox.
- Keep nulls:** An unchecked checkbox.
- Table lock:** A checked checkbox.
- Check constraints:** A checked checkbox.
- Rows per batch:** An empty text input field.
- Maximum insert commit size:** A text input field containing '2147483647'.
- View Existing:** A button at the bottom left of the main area.
- OK, Cancel, Help:** Buttons at the bottom right of the dialog.

2. Result

2.1. Query Result

The query result is provided in the Result.txt file.

2.2. Result in SSMS Screenshot



SQLQuery1.sql - DE...9DI2S\Gavriel (64)*

SELECT * FROM Result;

100 %

Results Messages

	SalesOrderID	OrderDate	Amount
136	49575	2013-01-31 00:00:00.000	2443.35
137	49576	2013-01-31 00:00:00.000	2181.56
138	49577	2013-01-31 00:00:00.000	2181.56
139	49578	2013-01-31 00:00:00.000	1000.43
140	49579	2013-01-31 00:00:00.000	2049.09
141	49580	2013-01-31 00:00:00.000	2049.09
142	49581	2013-02-01 00:00:00.000	2443.35
143	49582	2013-02-01 00:00:00.000	2181.56
144	49583	2013-02-01 00:00:00.000	2181.56
145	49584	2013-02-01 00:00:00.000	2443.35
146	49585	2013-02-01 00:00:00.000	2049.09
147	49586	2013-02-01 00:00:00.000	2071.41
148	49587	2013-02-01 00:00:00.000	2071.41
149	49588	2013-02-01 00:00:00.000	2071.41
150	49589	2013-02-01 00:00:00.000	2049.09
151	49590	2013-02-01 00:00:00.000	2049.09
152	49591	2013-02-01 00:00:00.000	2071.41
153	49594	2013-02-01 00:00:00.000	2181.56
154	49595	2013-02-01 00:00:00.000	1000.43
155	49596	2013-02-01 00:00:00.000	2181.56
156	49597	2013-02-01 00:00:00.000	2443.35
157	49598	2013-02-02 00:00:00.000	2181.56
158	49599	2013-02-02 00:00:00.000	2443.35
159	49600	2013-02-02 00:00:00.000	1000.43
160	49601	2013-02-02 00:00:00.000	1000.43

Query executed successfully. DESKTOP-S19DI2S\SQLEXPRESS ... DESKTOP-S19DI2S\Gavrie... AdventureWorks2022 00:00:00 46,555 rows

2.3. SQL Script to Create Result Table

```
CREATE TABLE Result (  
    SalesOrderID INT PRIMARY KEY,  
    OrderDate DATETIME,  
    Amount DECIMAL(10,2)  
);
```